

CENTRAL CALIFORNIA
EMERGENCY MEDICAL SERVICES
A Division of the Fresno County Department of Public Health

Manual	Emergency Medical Services Administrative Policies and Procedures	Policy Number 530.02
Subject	Paramedic Treatment Protocols GENERAL PROCEDURES	Page 1 of 23
References	Title 22, Division 9, Chapter 3.3 of the California Code of Regulations	Effective Fresno County: 01/15/82 Kings County: 04/10/89 Madera County: 06/15/85 Tulare County: 04/19/05

I. OVERVIEW OF HOW TO USE THE PROTOCOLS

This document contains the following:

- A. The standard of care and scope of practice for the paramedics in Fresno/Kings/Madera/Tulare Counties, and the procedures for the paramedics to follow.
- B. The treatment protocols for both medical and trauma emergencies. Each individual protocol outlines which treatment can be performed as a standing order and which treatment will need Base Hospital contact and/or Base Physician approval.
- C. The point where transport should be initiated for each protocol.
- D. Advanced Life Support (ALS) procedures which a Paramedic may perform without Base Hospital contact in the event of communications failure. ALS which a paramedic may perform in the event of communications failure is deemed in "lower case type." ALS orders written in "CAPITALIZED TYPE" may **NOT** be performed in the event of communications failure. Any delivery of ALS without Base Hospital contact due to communications failure must be reported according to EMS Policy #545 – Reporting Advanced Life Support without Base Hospital Contact.
- E. Procedures with an asterisk (*) may only be implemented by order of a Base Hospital Physician.
- F. Procedures without an asterisk may be implemented by order of a MICN without consulting a Base Hospital Physician.
- G. All blood pressure references are systolic.
- H. Paramedics are not allowed to switch protocols unless the patient needs treatment under an ACLS protocol (i.e., cardiac arrest protocols, PSVT, V-Tach, and Bradydysrhythmias). Protocols where a monitor is suggested allows for the paramedic to treat rhythm if appropriate. Switching protocols due to changes in patient condition should only occur after development of a new patient complaint that changes therapy as designated in the specific protocol or after Base Hospital consultation.

Approved By	Daniel J. Lynch	Revision
EMS Director	(Signature on File at EMS Agency)	07/01/2026
EMS Medical Director	Miranda Lewis, MD (Signature on File at EMS Agency)	

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II. ADVANCED LIFE SUPPORT BEFORE BASE HOSPITAL CONTACT

A Paramedic may initiate ALS prior to or without Base Hospital contact according to each specific treatment protocol. Each treatment protocol consists of a table divided into two parts. The top part is titled **STANDING ORDERS** and outlines the procedures for treating a patient that the paramedic is expected to perform, if indicated, prior to or without Base Hospital contact for a patient with that specific patient condition. All **STANDING ORDERS** are listed in sequential order for each condition, unless otherwise noted.

The bottom portion of the table outlined in bold is titled **BASE HOSPITAL ORDERS** and outlines the treatment procedures which require Base Hospital contact. Treatment which requires Base Hospital contact may be given by Base Hospital Physicians or MICNs. Procedures with an asterisk (*) may only be implemented by order of a Base Hospital Physician. Procedures without an asterisk may be implemented by order of a MICN.

III. FORMAT FOR BASE AND/OR RECEIVING HOSPITAL COMMUNICATIONS

The following formats will be used when transmitting the report of a patient's assessment, notification of therapy that has been completed and any additional therapy needed for treatment of a patient.

A. No Call-In Necessary

Patients that generally do not need a call-in to the Base Hospital:

1. Non-Stat (stable)
2. Patients that have been treated by standing orders and do not need additional therapy.

B. ETA Only Call-In

1. STAT/Non-STAT/Medical or Trauma/Code Blue.

NOTE: If requesting consultation, further orders, refusal of medical care and/or transportation, Paramedic must contact a Base Hospital using a standard format.

2. ETA notification must be provided when indicated for the "BIG FOUR" - backboard, restraints, active labor, or patients on oxygen.

C. Standard Call-In

Patients that require Standard Call-ins:

1. Refusal of medical care and/or transportation that require Base Hospital contact (EMS Policy #544 – Criteria for Base Hospital Contact).
2. Non-Stat or Stat calls where the patient needs additional therapy not allowed in standing orders.
3. Base Hospital requires additional information and asks for a standard call-in.
4. Paramedics switching or combining protocols when patient's complaint has not changed from original complaint (see Page 1 – H).
5. Patients requesting transport to hospitals on diversion.
5. Paramedics should call the Base Hospital any time they have questions or need help with interpretation of patient's condition.

- D. The severity of the patient's problem shall be identified with the following designators when hospital contact is made:

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1. “STAT” – Potentially life or limb-threatening conditions where the patient is unstable, in a rapidly changing status, or unstable as identified by the assessment and vital signs (e.g., Acute MI, etc.).
NOTE: Not necessarily Code 3, Refer to EMS Policy 548.
2. “NON-STAT” – Non-life or limb-threatening conditions, usually indicated by stable vital signs, including refusal of medical care and/or transportation that require Base Hospital contact by EMS Policy #544 – Criteria for Base Hospital Contact.
3. “Medical” or “Trauma” – Depending on the patient’s most severe problems. For example, alcohol intoxication causing a fall with severe head injury, would be designated, “STAT Trauma”.
4. “Code Blue” – When the patient is pulseless and/or non-breathing.

E. Call-in Format for Patients that Require Call-ins

1. ETA: Steps 1-5 Below (Base or Receiving Facility)
 - a. For use with Non-STAT patients with backboard, restraints, active labor, or patients on oxygen. If a receiving hospital does not answer the radio, there is no need to contact a Base to relay an ETA call-in. [May be made by either the Paramedic or EMT and can be received by any hospital personnel.]
 - b. STAT/ /Medical or Trauma/Code Blue. If short ETA and receiving hospital does not answer the radio, EMS personnel using **good judgement** may need to contact a Base Hospital or have dispatch contact the receiving hospital with an ETA of arriving EMS unit. [Can be received by any hospital personnel.]
2. Standard/ Diversion: Steps 1-13 Below (Base Only)
For use with STAT patients needing consultation, further orders, refusal of medical care and/or transportation, or for patients requesting transport to a hospital on diversion. [Must be made by the Paramedic (or EMT on BLS units only) and must be received by a MICN or Base Hospital Physician.]

F. At the initiation of call-ins, medics should identify the call as STAT/NON-STAT, Medical/Trauma, and ETA/Standard or Diversion (i.e., NON-STAT Medical ETA or STAT Medical Diversion). This will expedite communications and Base decisions.

S T A N D A R D / D I V E R S I O N	T R A U M A / V I T A L S I G N S	E T A / C O D E B L U E	<u>Steps 1-5:</u>
			1. Unit ID, Name
			2. ETA
			3. Age, Sex, Weight
			4. Chief Complaint, STAT/Non-STAT/Medical or Trauma/Code Blue
S T A N D A R D / D I V E R S I O N	T R A U M A / V I T A L S I G N S	E T A / C O D E B L U E	5. Reason for Notification – “BIG FOUR” (backboard, restraints, active labor, or patients requiring oxygen)
			<u>Steps 6-8:</u>
			6. GCS and Mechanism of Injury
			7. Vital Signs
			8. G-FAST Score and Last Known Normal Time (Suspected CVA only)
S T A N D A R D / D I V E R S I O N	T R A U M A / V I T A L S I G N S	E T A / C O D E B L U E	<u>Steps 9-14:</u>
			9. Physical Exam
			10. Standing Orders Completed or in Progress
			11. PMH
			12. Medications
S T A N D A R D / D I V E R S I O N	T R A U M A / V I T A L S I G N S	E T A / C O D E B L U E	13. Allergies
			14. Requesting Orders/ Refusal of Medical Care and/or Transportation

G. Multi-Casualty/Format (Base Hospital Only)
Refer to EMS Policy #620 “Multi-Casualty” for multiple patient incidents.

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H. Base Receiving Hospital Response Format

The MICN or Base Hospital Physician shall use a format for communicating with field personnel, which briefly highlights the Base Hospital response and key points of the prehospital patient report. This includes the patient profile, the radio operator's impression of the patient's primary problem and a description of the patient's vital signs. An example would be, "A 34-year-old male with crushing chest pain who is hypotensive."

If the call is ETA only to a receiving hospital, the receiving facility will simply identify the facility and personnel answering the radio and acknowledge the radio transmission.

I. Diversion Call-In Format

If a patient requests a hospital that is on diversion, a standard call-in is required. If the requested hospital is a Base Hospital, the call-in should go to that Base Hospital. If the patient requests transport to a receiving hospital (not a Base Hospital), the call-in should go to the appropriate Base (see note below). The Base Hospital will then notify the receiving hospital. (Refer to EMS Policy #547.1)

NOTE:

1. When the requested hospital is part of the Community Medical Centers (RMC, CCMC), the call-in should be made to RMC if possible; otherwise, geographic proximity (i.e., SAMC for Kaiser) should determine which Base is called.
2. To expedite patient destination decisions, these call-ins should be made as early as possible.
3. EMS Units should continue to prepare patients for transport and begin transport while waiting for Base Hospital response (i.e. diversions). Under **No** circumstances should EMS units wait on scene with stat patients for a Base Hospital decision for patient destination.

J. Hospital Contact and Call-in Listing

The following is a chart listing the hospitals and the types of calls they want from field. Also below the chart you will find detailed information for the described call-ins within the chart.

Call-In Format for Receiving Facilities						
ETA Call-in				Standard/Trauma Call-In		
HOSPITAL	ALL	BIG FOUR	STAT PATIENT	LABOR & DELIVERY (Complaints of Pregnancy)	ACUTE MI or CVA	TRAUMA
Adventist Health-Reedley		X				
Adventist Health-Selma		X				
Coalinga Regional Medical Center		X				
Community Medical Center-Clovis		X				
Kaiser Permanente		X				
Madera Community Hospital		X				
Veterans Administration	X					
Adventist Health-Hanford (B)		X		X		
Adventist Health-Tulare (B)		X	X	X		
Community Regional Medical Center (B)		X		X	X	X
Kaweah Health Medical Center (B)		X		X	X	X
Sierra View Medical Center (B)		X		X		
Saint Agnes Medical Center (B)			X	X	X	
Valley Children's Hospital (B)	X					X

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1. A standard call-in is required for patients that list the following on a cardiac monitor:
 - *** ACUTE MI *** (Zoll Monitor E Series)
 - ***STEMI*** (Zoll Monitor X Series))
 - ***ACUTE MI SUSPECTED*** (Physio-Control Monitor LifePak 12)
 - ***MEETS ST ELEVATION MI CRITERIA*** (Physio-Control Monitor LifePak 15)

Several hospitals are using prehospital field information from the call-in to determine the activation of stroke teams or cardiac catheterization lab teams.

NOTE: If a patient refuses one of the above destinations, an early ETA call-in should occur to the receiving facility requested by the patient.

2. A standard call-in is required for patients with complaints of pregnancy. Several hospitals are using prehospital field information from the call-in to determine patient destination within their facility (i.e., emergency department or labor and delivery department). EMS personnel are to go to the emergency department unless redirected by the MICN or Base Hospital Physician. EMS personnel should never assume that a pregnant patient goes directly to labor and delivery.

3. Stat Trauma Call-in

- Unit ID
- Name (Paramedic or EMT)
- ETA
- Age
- Chief Complaint
- Mechanism of Injury
- GCS
- Vital Signs

NOTE: (B) signifies a Base Hospital.

IV. LARYNGOSCOPY IN COMPLETE AIRWAY OBSTRUCTION BY FOREIGN BODY

If manual airway maneuvers (BLS Treatment Protocols, EMS Policy #510.12) fail to dislodge a foreign body causing complete airway obstruction (according to protocol), the Paramedic shall visualize the airway with laryngoscope, and if the object is visible, remove obstruction with Magill forceps. Attempt to remove foreign body with laryngoscopy for up to one minute. If unsuccessful, repeat ventilation attempts, and manual airway maneuvers, and initiate transtracheal jet insufflation according to protocol.

V. TRANSTRACHEAL JET INSUFFLATION

A. Indication:

1. Complete airway obstruction not relieved by manual procedures and airway visualization with laryngoscope.
2. Inability to intubate and inability to successfully ventilate using BVM ventilation.

B. Procedure:

1. Locate cricothyroid membrane.
2. Insert 10 gauge IV catheter through the membrane at a 45° angle, directed toward the feet. Aspirate for air return with a syringe to check placement. Remove needle.
3. Stabilize catheter securely to neck.
4. Attach pediatric anesthesia adapter to catheter.

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5. Supply 100% oxygen to anesthesia adapter via oxygen-powered breathing device.

NOTE: In children under 12 years of age, ventilate with Bag-Valve-Catheter with 100% oxygen. If unable to ventilate, use oxygen-powered breathing device.

6. Check for proper placement in the following order:
 - a. Assess chest rise.
 - b. Check absence of gastric sounds.
 - c. Check adequacy of breath sounds.
 - d. Assess for complications, including subcutaneous air.
 - e. Reassess placement every time patient is moved. Sometimes proper placement is difficult to assess. Do not just rely on the indicators listed above. Continual clinical reassessment for adequate oxygenation is essential.

C. SURGICAL CRICOTHYROTOMY IS NOT A LOCALLY APPROVED PARAMEDIC SKILL.

VI. NEEDLE THORACOSTOMY (CHEST DECOMPRESSION)

- A.** Indication: Will be identified in each individual protocol.

Signs and symptoms of Tension Pneumothorax, including **all of the following:**

1. Severe Respiratory Distress (as evidenced by apnea, severe dyspnea with tachypnea, oxygen saturation less than 90% for greater than 30 sec. (if utilized), or difficulty in bagging).
2. Lateralizing Exam (decreased breath sounds on one side, or tracheal deviation away from the tension, or asymmetric chest wall rise).
3. Hemodynamic Compromise (BP less than 90).

- B.** Procedure:

1. Use a 10 gauge IV catheter at least 3¼ inches long for an adult patient and 14 gauge catheter at least 1¼ inches long for pediatric patient.
2. The site preference for needle thoracostomy is:
 - a. Primary Site – mid-axillary at the fifth intercostal space (approximately nipple level) on the side of decreased breath sounds, or
 - b. Secondary Site – above the third rib (second intercostal space), slightly lateral to the mid-clavicular line (approximately at the level of the angle of Louis) on the side of decreased breath sounds.
3. When air returns, advance the catheter and remove the needle.
4. Attach a one-way valve to the catheter hub (if spontaneous respirations are present).
5. Stabilize the catheter securely to the chest.
6. Reassess the patient, including breath sounds and vital signs every time the patient is moved.

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VII. I-GEL SUPRAGLOTTIC AIRWAY

A. Indications:

The i-gel may be used as an advanced airway alternative to endotracheal intubation. It is performed only on a patient who meets all of the following:

1. Unconscious (no purposeful movement), with an absent gag reflex.
2. Apneic or agonal respirations less than 8.

B. Contraindications:

1. Intact gag reflex.
2. Trismus or inability to open the mouth.
3. Complete airway obstruction (i.e. Foreign body, severe edema, tumor).
4. Laryngectomy or tracheal stoma.
5. Caustic ingestion or burns involving the airway.
6. Significant oropharyngeal trauma.

C. Special Considerations

1. Chest compressions should never be interrupted for placement of the i-gel in the medical arrest patient.
2. No medications are to be administered through the supraglottic airway.
3. The pediatric i-gel packaging does not come equipped with a securing strap. A separately packaged securing strap must be used.
4. I-gel is not approved for use in neonates.

D. Procedure:

1. Airway obstruction is an absolute contraindication to the use of a supraglottic airway. Prior to placement of the supraglottic airway, confirm that the patient can be ventilated with a bag valve mask.
2. Estimate the patient's weight and select appropriate size i-gel.

Tube Size	Patient Size	Weight
1.5	Infant	5-12 kg
2	Small Child	12-25 kg
2.5	Large Child	25-35 kg
3	Small Adult	30-60 kg
4	Medium Adult	50-90 kg
5	Large Adult	90+ kg

3. Prior Suction airway as needed. Remove dentures, if present
4. Lubricate the back of the supraglottic airway with water-based lubricant taking special care to avoid the lumen of the tube.
5. Place the head in a sniffing or neutral position. Maintain spinal motion restriction, if indicated.

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6. Introduce the device into the mouth and advance the tip of the device along the hard and soft palate with a continuous but gentle pressure.
7. A slight give-way resistance may be felt during insertion as the device passes into the oropharynx. It is important to continue to insert the device until definitive resistance is felt. Do not apply excessive force on the device during insertion.
8. The i-gel integral bite block has a horizontal line to indicate optimal position of the tube in relation to the teeth.
9. Confirm placement with all of the following:
 - a. Auscultate breath sounds.
 - b. Evaluate for chest rise.
 - c. Continuous waveform capnography.
10. If there is any concern about proper placement, remove the device and ventilate with BVM for one minute before second attempt. Do not make more than two attempts to establish and i-gel.
11. Secure the tube using the manufacturer included strap and ventilate with a BVM and 100% O₂.
12. The pediatric i-gel must be secured with a separately packaged strap. There are no hooks for securing the strap on the pediatric i-gel. Instead, a hole on each side of the strap should be stretched over the BVM connector of the i-gel.
13. The device position should be reevaluated frequently and at minimum:
 - a. Anytime the patient is moved.
 - b. When there is a change in patient status.
 - c. When there is difficulty ventilating the patient.
14. If there is a change in patient status after placement of the supraglottic device and continued ventilation using a supraglottic airway is no longer possible (i.e. patient combativeness, ineffective ventilation, intact gag reflex), the device should be removed. Be prepared to suction the patient on removal and assist ventilations with BVM as appropriate.

VIII. ENDOTRACHEAL INTUBATION

A. Indications:

Intubation may be completed prior to Base Hospital Contact if patient meets any of the following:

1. Patients >14 years of age in cardiac or respiratory arrest; or,
NOTE: In-line cervical immobilization must be used for patients with a possible neck injury.
2. Spontaneously breathing patients with GCS of 8 or less, who are unable to maintain their own airway or without gag reflex; or,
3. Spontaneously breathing patients with a GCS of 8 or less who have a respiratory rate less than 8 per minute.

B. Contraindications:

1. The patient in respiratory arrest from a suspected narcotic overdose should not be intubated until AFTER administration of Naloxone.

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2. PEDIATRIC ENDOTRACHEAL INTUBATION IS **NOT** AN APPROVED SKILL FOR PARAMEDICS.
 3. NASAL ENDOTRACHEAL INTUBATION IS **NOT** A LOCALLY APPROVED SKILL FOR PARAMEDICS.
 4. THE PATIENT HAS A DO-NOT-RESUSCITATE ORDER (DNR) – EMS POLICY #564.
- C. Procedure:
1. Provide in-line cervical immobilization for patients with a possible neck injury.
 2. Have suction equipment immediately available.
 3. Prior to performing endotracheal intubation, hyperventilate the patient with 100% oxygen for a minimum of one minute, if possible.
 4. Do not interrupt ventilations for more than 20 seconds while performing endotracheal intubation. If unable to insert and ventilate within 20 seconds, stop, hyperventilate and reattempt.
 5. If unsuccessful, ventilate for one minute before trying again.
 6. Do not make more than two (2) attempts total per patient to perform endotracheal intubation. An attempt shall be defined as any cessation of ventilations in order to perform laryngoscopy. If either unsuccessful after 2 attempts, or intubation not felt possible, insert an i-gel.
 7. Check for proper placement in the following order:
 - a. Check adequacy of breath sounds in each hemithorax.
 - b. Check absence of gastric sounds.
 - c. End Tidal CO₂ Detector (for patients that are mechanically ventilated, with or without a pulse, ETT/i-gel/trach, and BVM).

NOTE: THIS DEVICE IS TO BE USED AS AN ADJUNCT TO ASSESS TRACHEAL INTUBATION. ITS PURPOSE IS NOT TO ELIMINATE CLINICAL JUDGMENT.

NOTE: Agencies with Capnography may utilize in place of End Tidal CO₂ Detector Device.
 - d. Esophageal Detector Device (only patients without a pulse).

NOTE: Do not use in children less than 5 years of age or less than 20 kg.

NOTE: THIS DEVICE IS TO BE USED AS AN ADJUNCT TO ASSESS TRACHEAL INTUBATION. ITS PURPOSE IS NOT TO ELIMINATE CLINICAL JUDGMENT.

NOTE: Agencies with Capnography may utilize after use of the EDD for continuous monitoring of ET placement.
 - e. Assess chest rise and ease of ventilation.
 - f. Positive fogging.
 - g. Assess for complications.

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- h. Reassess placement of ET tube every time the patient is moved or ET tube manipulated.
- i. In most adult patients, a properly placed ET tube will have the 22 cm mark at the level of the teeth.
8. Secure the ET tube as soon as possible.
9. After endotracheal intubation, ventilate with bag-valve and 15 liters/minute of oxygen with reservoir.
- D. Medication Administration via the Endotracheal Route:
 1. Indication:

Medications via the endotracheal tube route should be given only when an IV/IO cannot be established. The ability to administer medications by the ET tube route should not preclude attempts to start IV/IO.
 2. Procedure:

Hyperventilate the patient with 100% oxygen. Hold chest compressions briefly while drugs are being administered down the endotracheal tube. If volume of medication is less than 1.5 ml, then flush with 2 ml sterile normal saline. Ventilate four times in rapid succession after the administration of any medication and its accompanying saline.
 3. Give only the following drugs via the endotracheal route:
 - Epinephrine
 - Atropine
 - Naloxone
 - Midazolam

IX. OXYGEN THERAPY

A. Indications:

Oxygen therapy is required only for patients with the following conditions as defined in protocol:

1. Acutely altered mental status or any acute neurological symptoms (i.e., seizure, syncope, stroke, etc.).
2. Respiratory distress, cyanosis, altered respiratory rate, inhalation injuries, or exposures.
3. Any chest pain of possible cardiac or pulmonary etiology.
4. Shock.
5. Abnormal heart rate.
6. Significant multiple system trauma or patient meeting regional trauma center triage criteria.

NOTE: This should not be based on mechanism of injury alone, but by the patient's condition and vital signs.
7. Any other condition specifically covered in the BLS or Paramedic protocols.
8. Patients shall be provided oxygen in accordance with protocols, patient signs and symptoms, or when a patient's pulse oximetry reading is 93% or less.

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B. Oxygen Delivery and Dose:

1. Administer oxygen to spontaneously breathing patients according to specific protocols using either:
 - a. Low flow – Nasal cannula (6 liters/min).
 - b. High flow – Non-rebreathing oxygen mask (15 L/min). (Be sure to keep reservoir bag inflated.)
2. If patient has a history of COPD, start oxygen at 2 L/min by nasal cannula. If cyanotic, gradually increase oxygen flow until cyanosis clears. If still cyanotic on 6 L/min by nasal cannula change to 15 L/min by non-rebreathing mask.

 If the patient is on home oxygen and is chronically cyanotic, administer the patient's normal oxygen dosage and contact the Base Hospital regarding increasing the oxygen flow. Prepare to assist ventilations with bag-valve-mask, since oxygen may cause sleepiness and hypoventilation.
3. Use bag-valve-mask with supplemental oxygen (15 L/min) and reservoir or oxygen-powered breathing device when:
 - a. Patient is not breathing.
 - b. Patient's breathing is too shallow to ventilate adequately.
 - c. Patients with a GCS of 8 or less should have their ventilations assisted.

NOTE: If a patient has a GCS of 8 or less, especially if they meet criteria for intubation and are not intubated, that patient should have their ventilations assisted by BVM or oxygen-powered breathing devices.

NOTE: Should not utilize an oxygen-powered breathing device on children less than 5 years of age except as allowed with transtracheal jet insufflation.

X. VASCULAR ACCESS

A. Establishing IV Access

Consider establishing IV access in any patient when there is a reasonable chance that the patient's condition may deteriorate enroute.

B. Saline Lock IV Access

Initiate a saline lock IV access in all patients who require vascular access based upon the specific treatment protocol, but who need no immediate medication administration or fluid administration (e.g., IV TKO and transport). Pre-existing saline lock IVs on the extremities may be used for fluid or medication administration.

If medication administration is needed due to a change in patient status, inject medication into the lock, and flush with saline. If multiple doses of medication are anticipated, consider attaching an IV bag with tubing to the saline lock with a transfer needle using sterile technique.

- C. IV Access – Start one or two IVs of Lactated Ringer's solution in patients who require vascular access according to specific protocol.
- D. External Jugular Venipuncture – Paramedics may use the external jugular route for IV insertion, if unable to establish an IV via other peripheral routes, and the IV is essential for patient care.
- E. Pre-existing Vascular Access – If unable to establish a peripheral IV, and the patient needs fluids and/or medications, the Paramedic may use pre-existing vascular access. Refer to EMS Policy #565 (Maintenance of

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Patient Controlled Drugs) for patient controlled therapy. This can be used as the primary source in a code situation.

- F. Intraosseous Access – The intraosseous (IO) access route is for fluid and medication administration if unable to establish an IV, and access is essential for patient care.

1. Indications:

- a. In critically ill pediatric patients, where IV access is essential, and when one initial IV attempt is unsuccessful, or when no vein is immediately apparent, start intraosseous access without further IV attempts. If patient is awake and alert, contact Base Hospital prior to inserting IO line. (Volutrol will not work on IO lines; therefore, use 60cc syringe to give weight appropriate boluses and flush all medications with 10cc of LR or NS.)
- b. In adult or pediatric cardiac arrest patients, IV or IO access is acceptable. If ET tube is available, administer medications via the endotracheal route if IV/IO access cannot be obtained.
- c. Other patients may have intraosseous access started only by Base Hospital order.

All prehospital medications administered via the intravenous route may be administered through intraosseous access according to specific protocols.

2. Contraindications:

- a. Extremity cannot be used if fracture is present.
- b. Avoid areas where a burn or cellulitis is present.

3. Procedure:

- a. The site preference for intraosseous infusion is:

(1) Pediatrics

First choice – the flat, medial surface of the anterior tibia, 1-2 cm below the tibial tuberosity.

Second choice – distal anterior femur, midline 3 cm above the patella.

(2) Adults

2-3 cm proximal to the medial malleolus and direct it slightly upward.

Humeral head

4. Place intraosseous needle (with stylet) either perpendicular to the bone or 45 degrees away from the nearest growth plate. Apply firm downward pressure using a rotary motion to enter the bone marrow. There will be a sudden decrease in resistance when entering the marrow.
5. When the needle is standing without support, remove the stylet and attempt to aspirate marrow.
6. Although not always present, visible aspiration of marrow content (i.e., dark blood) will confirm proper placement. Easy flow of IV solution alone does not confirm correct placement. IV fluid may not drip to gravity; pressure may be required by use of blood pressure cuff or 60cc syringe. The best indicator of proper placement is the absence of tissue swelling around the site during fluid infusion.
7. If properly placed, the needle needs no securing. It will be solidly lodged in the bone. The IV tubing should be taped.

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G. IV Fluid Administration Procedure

1. For patients not requiring fluid administration (according to protocol), maintain IV at TKO.
2. For patients with traumatic cardiac arrest or who require rapid volume replacement (according to protocol), administer lactated ringers wide open until systolic BP greater than 100 or until a total of two liters infused. Contact Base Hospital for fluid orders once systolic BP greater than 100 or two liters is infused.
3. For patients who require a fluid challenge according to protocols, administer the fluid as follows:
 - a. Bolus until 500 ml LR is infused or the amount ordered by the Base is infused, then return IV to TKO.
 - b. Check vital signs and lung exam after each 500 ml LR bolus or the amount the Base ordered.
 - c. Repeat bolus of 500 ml LR or amount ordered by the Base if patient still meets protocol criteria.
 - d. If signs or symptoms of shock persist after a total of two 500 ml LR boluses, contact the Base Hospital for fluid orders. Consider another bolus.
4. If signs of pulmonary edema or a new onset of dyspnea develop during IV fluid administration: Slow IVs to TKO. Contact the Base Hospital for fluid orders.

XI. SPINAL MOTION RESTRICTION

A. Goal:

The goal of Spinal Motion Restriction (SMR) is to prevent secondary injury and worsening morbidity in patients with unstable spinal injury, reserve full spinal precaution use for high-risk patients, and avoid complications associated with full spinal immobilization.

B. Indications for SMR in blunt trauma with suspected spinal injury include:

1. Acutely altered level of consciousness (GCS<15) or evidence of intoxication such that their appreciation of pain or ability to communicate is impaired in the setting of trauma.
2. Significant midline neck or back pain/tenderness in the setting of trauma.
3. Focal neurologic deficits (numbness/paresthesia or weakness) in the setting of trauma.
4. Anatomic deformity of the spine secondary to trauma.
5. Distracting circumstances/injury that impairs the patient's ability to contribute to a reliable exam in the setting of trauma (i.e. long bone fracture, significant pelvic injury, severe head injury).

If the above criteria are not met, but there is clinical concern for spinal injury, the patient should be placed in SMR.

C. Contraindications:

1. Isolated penetrating trauma.
2. Facial/oral bleeding or uncontrolled vomiting such that the airway cannot be controlled with SMR in place.
3. Uncontrolled hemorrhage in the face or neck area that cannot be managed with SMR in place.

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D. Procedure:

1. Spinal motion restriction does not take precedence over airway, respiratory, and cardiovascular stabilization in the critical trauma patient.
2. Assess for possibility of spinal injury by performing history and exam:
 - a. Check wrist/finger flexion (both hands), plantar/dorsiflexion (both feet).
 - b. Check gross sensation in all extremities and evaluate for paresthesia.
 - c. Palpate the midline cervical, thoracic, lumbar spine to evaluate for tenderness.
3. Apply appropriately sized cervical collar while a second provider maintains inline stabilization of the cervical spine while avoiding traction.
4. Moving the head to a neutral inline position is contraindicated if:
 - a. There is pain upon starting movement.
 - b. There is muscle spasm or resistance to movement.
 - c. The head is held rigidly to one side.
5. The remainder of the spine should be stabilized by keeping the head, neck, torso in alignment. This can be accomplished by securing the patient supine on a long back board, scoop stretcher, breakaway flat, vacuum mattress, or on the ambulance gurney. Lateral head support with blocks or towel rolls should be used when possible.
6. Allow ambulatory patients to sit on the stretcher and then lie flat (no standing take down).
7. If there is prolonged transport time and patient is alert and cooperative, the extrication device (i.e. long back board) may be removed once the patient is safely positioned on the ambulance gurney provided there are adequate personnel available to minimize unnecessary movement of the spine during removal. Do not delay transport for removal of the long back board in the critical trauma patient. If the long back board is removed, SMR must still be maintained with the patient firmly secured to the gurney with seatbelts and cervical collar in place. A moving device should be placed under the patient to facilitate movement off the EMS gurney on hospital arrival.
8. Reassess and document sensation and motor function following SMR.
9. If elevation of the head is required, the device used to stabilize the spine should be elevated at the head while maintaining alignment of the neck and torso (elevation of head no greater than 30 degrees). SMR cannot be properly performed with a patient in a sitting position or high fowlers.

E. Pediatric patients:

1. The approach to spinal motion restriction is similar for pediatric patients. The decision to apply SMR should not be made based solely on age.
2. Indications for SMR in blunt trauma include:
 - a. Complaint of neck pain, inability to move the neck, or midline tenderness in the setting of trauma.
 - b. Focal neurologic deficit (numbness/paresthesia or weakness) in the setting of trauma.
 - c. Altered mental status in the setting of trauma.

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- d. Involvement in high-risk motor vehicle collision, high impact diving injury, or substantial torso injury.
3. Appropriate patients may be secured to the gurney in their car seat. If an appropriately sized collar cannot be safely fitted, the spine may be stabilized using a combination of towels, soft collar, or lateral head support.

F. Special Considerations:

1. SMR does not take precedent over airway, respiratory, and cardiovascular stabilization. Rapid extrication and transport remain top priority for the critical trauma patient.
2. SMR in penetrating trauma is associated with increased mortality. SMR should not be used in isolated penetrating trauma.
3. A poorly fitting cervical collar is worse than no collar at all. Distraction or hyperextension of the cervical spine due to a poorly fitting cervical collar may lead to worsening injury.
4. Patients with severe kyphosis, size constraints, or other conditions causing anatomic deformity of the spine (i.e. ankylosing spondylitis) may be stabilized using a combination of lateral head support with blocks or towel rolls, blankets, soft collar, vacuum splint or other devices if an appropriately sized cervical collar cannot be fitted.
5. Cervical immobilization is not necessary for patients complaining of isolated lumbar pain/tenderness who do not have a neurologic deficit, intoxication, altered mental status, or distracting injury present. Immobilization can be accomplished by securing the patient supine on a long back board, scoop stretcher, breakaway flat, vacuum mattress, or on the ambulance gurney.
6. If attempting to apply SMR to a combative patient would cause further detriment, abort the procedure and document in the ePCR. Notify ED staff on arrival regarding indications for SMR but the inability to apply due to combative patient.

XII. TRACHEOSTOMY CARE/SUCTIONING

A. Indications:

1. Any patients with respiratory distress and a tracheostomy.

B. Contraindications:

1. None

C. Procedure:

1. Assess Problem
 - a. Obstruction (Partial/Complete)
 - Excessive secretions
 - Dried secretions
 - Obstructed inner cannula
 - Swelling/Infection
 - Foreign Body
 - Bleeding (Rare)
 - b. Dislodgment
 - Sutures or tapes loose
 - Trach tube out of tracheostomy

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2. Type of Tube:
 - a. Single or double lumen – Both types have balloons similar to ET tubes. Double lumen tubes have an inner cannula that is removable.
3. Treatment:
 - a. 100% oxygen by BVM or mask to trach based on patient ability to ventilate.
 - b. Remove inner cannula if double lumen tube (rinse with sterile saline).
 - c. Suctioning (use sterile gloves as necessary).
 - Preoxygenate.
 - Irrigate with 3cc NS through trach (i.e., saline flush, irrigation syringe).
 - Tell patient to exhale.
 - Insert suction catheter gently until resistance is felt.
 - Tell patient to cough/exhale.
 - Suction during withdrawal of catheter.
4. Contact Base Hospital for medical direction if:
 - a. Patient not improving after above treatments.
 - b. Patient develops subcutaneous emphysema.
 - c. Patient is bleeding from trach.
 - d. Trach tube is dislodged.
 - e. You have any questions.

Irrigate with 3cc NS through trach (i.e., saline flush, irrigation syringe).

NOTE: Do not delay transport. STAT transport, contact base hospital enroute to closest appropriate facility.

XIII. TRANSCUTANEOUS PACING (TCP)

A. Indications:

1. Patients with serious signs and symptoms related to bradycardia (HR less than 60 beats/minutes) who have failed to respond to pharmacological therapy. “Hemodynamically significant” may be defined by severe chest pain, severe shortness of breath, pulmonary edema, acutely altered mental status, or shock (BP less than 90).
 - a. In an **adult**, pacing should be attempted if the patient has a persistent, symptomatic bradycardia after a total of 1 mg of Atropine has been administered.
 - b. In a **child**, oxygenation, ventilation, epinephrine, and atropine all should precede pacing.
2. The pacer may also be used on the order of a physician who is initiating an interfacility transfer or on the order of a Base Hospital Physician in a case where the paramedic has requested on-line medical control.

B. Contraindications:

1. Patients with no signs/symptoms related to bradycardiac rhythm.

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C. Procedure:

1. Consider administering Fentanyl and/or Midazolam (per Policy #530.10). If unable to start IV, consider administering IM.
2. Place pads on the patient's chest and back.
3. Set initial TCP rate at 80 beats per minute (bpm).
4. Begin output at 0 milliamps (mA). Increase by 10mA until capture/pulses are noted. Once capture is confirmed, continue pacing at a slightly higher output level (10%).
5. If capture is maintained but the patient remains symptomatic of inadequate tissue perfusion (B/P less than 90 systolic, altered level of consciousness), consider increasing **the rate** by 10 bpm until 100 bpm is reached.

D. Troubleshooting:

1. Make sure the pads are properly placed and have good contact with the skin.
2. Check the batteries of the pacer.
3. Use adequate energy to capture the rhythm.
4. Use adequate analgesia and sedation to minimize patient discomfort.

XIV. 12-LEAD ECG

A. Indications:

1. Coronary ischemic chest discomfort
2. Shortness of breath with pulmonary edema (if acute onset of symptoms)
3. Unexplained hypotension (in presumed cardiogenic shock)
4. Return of spontaneous circulation (after any ACLS treatment for arrest)

NOTE: All other 12-LEAD ECG applications **require Base Hospital approval.**

B. Contraindications:

1. None

C. Procedure:

1. Apply 12-lead ECG per manufacturer's instructions.
2. Use limited to that described in Central California Policies and Procedures.

XV. CONTINUOUS POSITIVE AIRWAY PRESSURE (CPAP)

A. Indications:

1. Severe Shortness of Breath with Bronchospasm (Including COPD and Asthma)
2. Severe Shortness of Breath with Pulmonary Edema (Including Congestive Heart Failure)

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3. Allergic Reactions with severe bronchospasm.
4. Conscious, breathing spontaneously, and able to follow commands
- B. Contraindications:
 1. Pediatric patients (14 years old or less)
 2. Actively vomiting
 3. Hypotensive (systolic blood pressure less 90)
 4. Suspected of having a pneumothorax
 5. An inability to achieve a good facial seal with the CPAP mask.
- C. Procedure:
 1. Do not delay medication administration to apply CPAP.
 2. The patient must be continuously monitored for development of respiratory failure or vomiting. If either occurs, remove the CPAP circuit, clear the airway as necessary to prevent any aspiration, and provide respiratory assistance with either BVM or other advanced airway adjunct.
 3. CPAP will be delivered at a continuous pressure of 5 up to 10 cm H₂O utilizing 100% oxygen.
 - a. Start CPAP at 10 cm H₂O and decrease if possible.
 - b. Start oxygen at 100% and titrate for oxygen saturation greater than 95% if possible.
 4. CPAP may introduce transient hypotension via decreased venous return secondary to elevated intrathoracic pressure.
 - a. If systolic blood pressure falls to less than 80 mmHG, remove CPAP.
 - b. If systolic blood pressure falls between 80-100 mmHG, decrease CPAP to 5 cm H₂O if possible.

XVI. VENTRICULAR ASSIST DEVICES

- A. Procedure:
 1. Assess the patient – There may be no palpable pulse, therefore, utilize other parameters for patient assessment (e.g., mental status, skin signs, capillary refill, and ETCO₂).
 2. Assess the device.
 - a. Device information and VAD Coordinator contact number may be on the device.
 - b. If caregiver is present, yield to their advice.
 - c. For continuous flow devices (no palpable pulse), auscultate over the left upper quadrant of the abdomen or over the heart and listen for the “hum” of the device.
 - d. Determine if the device has power (If the device has power, it does not necessarily mean it is working properly). A malfunctioning pump should beep or may have a blinking or solid red light.
 - e. Check the device for secure connections.

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3. If the patient's condition appears to be related to their VAD, and it is safe and reasonable, it is preferred to transport the patient to the facility, within the CCEMSA region, in which their VAD Coordinator works. If the patient condition warrants transport to a closer or more appropriate hospital (i.e. trauma or burn center) as directed in Policy 547, follow patient destination criteria as stated in policy.

B. Special Considerations

1. Prehospital providers should rely upon the patient's level of consciousness, skin signs, capillary refill, respirations, and ET CO_2 to make clinical decisions. Pulse-oximetry and blood pressures will not be obtainable in patients with a second-generation VAD device.
2. Patients may be cardioverted or defibrillated if symptomatic.
3. There are no absolute medication contraindications for VAD patients.
4. Chest compressions are not contraindicated and can be performed, but only if you are certain the pump is not working and/or there is not any flow through the VAD. Mechanical CPR devices (e.g. Lucas Device) are **CONTRAINDICATED**.
5. Treatment should follow appropriate treatment protocols.
6. Contact the Base Hospital as needed.
7. If possible, the patient's family member or caregiver should accompany the patient to the hospital. All related VAD equipment including spare batteries should be transported with the patient.
8. In arrest situations, determine if a POLST/DNR or Advanced Directive is available.

XVII. CONTINUOUS WAVEFORM CAPNOGRAPHY

A. Indications:

1. Confirmation of proper endotracheal tube and i-gel placement.
2. Continuous monitoring of patients with an advanced airway.
3. Monitoring of resuscitation efforts during cardiac arrest in patients with advanced airway.

B. Contraindications:

1. None

C. Procedure:

Confirmation and monitoring of advanced airway:

1. Continuous waveform capnography must be used for all patients receiving advanced airway management (endotracheal tube and i-gel placement).
2. Attach the capnography device to the monitor prior to applying the device to the patient. This ensures that the device properly zeroes to ensure an accurate reading.
3. Attach the capnography device to the bag valve mask prior to intubation attempt.
4. The capnography device should be used to confirm placement immediately following insertion of the advanced airway. Additionally, intubation should be confirmed through direct visualization, confirmation of equal breath sounds, and absence of gastric sounds.

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5. Proper placement is confirmed by presence of normal capnography waveform with ventilation. A normal ETCO₂ has a square appearance. A normal value is between 35-45 mmHg. Patients in cardiac arrest or those with poor perfusion may have low ETCO₂ readings (< 10 mmHg) however, there should still be a normal square shaped tracing.
6. An ETCO₂ value less than 10 mmHg without typical appearance of the waveform is indicative of esophageal placement or misplacement. The endotracheal tube or i-gel must be immediately removed.
7. A sudden increase in the ETCO₂ value during cardiac arrest is suggestive of ROSC.
8. Continuously monitor the waveform ETCO₂ tracing. The monitor tracing showing the initial waveform must be attached to the PCR. If the waveform cannot be attached to the PCR due to technologic failure, the numeric ETCO₂ value and presence of waveform must be documented in the PCR at the following time intervals:
 - a. Immediately after placement of an advanced airway.
 - b. After any patient movement.
 - c. Every 5 minutes during transport.
 - d. On turnover of care to the emergency department.
9. An appropriate ETCO₂ waveform confirms the proper advanced airway placement (ETT and i-gel) and may be used in lieu of physician signature for advanced airway confirmation **provided that the ETCO₂ waveform is attached to the PCR.**

XVIII. MECHANICAL CPR DEVICES

Manual chest compressions are the standard of care for patients in cardiopulmonary arrest. Studies have shown no mortality benefit to support the use of mechanical CPR devices over high-quality manual chest compressions. However, there are situations where manual CPR is challenging or dangerous for the prehospital provider and mechanical chest compressions are preferred.

- A. Indications:
 1. Patients being transported with ongoing CPR
 2. Prolonged resuscitation (> 10 minutes) to prevent rescuer fatigue or when limited rescuers are available
 3. Prophylactic application prior to transport in patients with ROSC in case of rearrest. Device should only be activated in the event of rearrest
 4. Cardiac arrest patients located in a confined space where manual CPR and rapid extrication are not possible
- B. Contraindications:
 1. Patient is too small for the device to be applied per manufacturer instructions
 2. Patient is too large for the device to be applied per manufacturer instructions
 3. Age and weight restrictions per manufacturer instructions
 4. Device cannot be appropriately positioned on the chest
 5. Cardiac arrest due to trauma

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6. Patients with ventricular assist devices

C. Procedure:

1. Manual CPR should be performed immediately on patient arrival. Do not delay initiation of chest compressions to place the mechanical CPR device.
2. Ensure the chest is exposed prior to placement of the device.
3. The CPR feedback device must be removed prior to application of the mechanical CPR device, unless otherwise specified in manufacturer instructions.
4. Minimize interruptions in compressions when applying the device. The device should be applied in a choreographed stepwise fashion with manual chest compressions delivered between steps.
5. Follow device specific manufacturer instructions for application and operation.
6. Ensure appropriate defibrillator pad placement and reassess pad placement with each rhythm check.

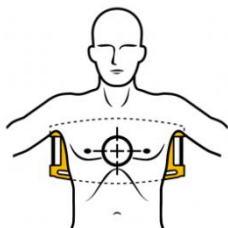
D. Special Considerations:

The following devices are approved for use in the CCEMSA region. Any additional devices shall be approved by the EMS Agency prior to implementation.

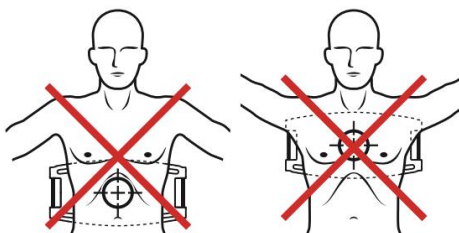
LUCAS:

1. It is critical that the suction cup is placed in the appropriate position indicated below and continually reassessed during transport.

Correct placement:



Incorrect placement:



2. The neck stabilization strap must be applied prior to patient movement. Secure the arms to the device using the device straps. Additionally, a permanent marker should be used to mark the upper and lower edges of the suction cup on the patient's chest. These markings will be referenced as landmarks to ensure the suction cup is in appropriate position throughout transport.
3. If the suction cup position migrates during transport, it must be immediately corrected. Misplacement of the device may lead to inadequate circulation and internal injury.
4. Defibrillation may safely be applied while the LUCAS device is providing compressions.

Zoll AutoPulse

1. Ensure that the device is appropriately positioned according to the manufacturer instructions.
2. The device should be continually monitored during transport to prevent migration.

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3. Ensure shoulder restraint should be applied prior to transport to keep the patient properly aligned on the AutoPulse platform.

XIX. NEEDLE CRICOTHYROTOMY/QUICKTRACH

A. Indications:

1. Complete airway obstruction not relieved by manual procedures or airway visualization with laryngoscope.
2. Inability to ventilate with bag valve mask ventilation, supraglottic airway, and intubation.

B. Contraindications:

1. **QuickTrach is contraindicated in pediatric patients <2 years or <10 kg.** Needle cricothyrotomy should be used in this age group.
2. Inability to identify anatomic landmarks.
3. Ability to ventilate or oxygenate by any other means.

C. Needle Cricothyrotomy

1. Place the patient in a supine position with support under the shoulder and mild hyperextension of the head. If spinal precautions are indication, maintain the neck in a neutral position.
2. Locate the cricothyroid membrane.
3. Prepare the skin of the anterior neck with antiseptic solution.
4. Stabilize the cricoid cartilage with the nondominant hand.
5. Attach a saline flush with 2-3 ml saline or a syringe to a 10-gauge IV catheter. Insert the needle through the cricothyroid membrane at a 45-degree angle, directed toward the feet.
6. Aspirate for air return as the catheter is inserted. Once air return is obtained, hold the needle in place advance the catheter over the needle. Remove the needle once the catheter is advanced to an appropriate depth.
7. Manually hold and stabilize the catheter. Attach the pediatric anesthesia adapter.
8. Supply 100% oxygen to anesthesia adapter via oxygen powered breathing device.
Note: The oxygen powered breathing device should not be used in pediatric patients due to risk of barotrauma. Ventilate pediatric patients with a bag valve mask.
9. Confirm placement immediately with continuous waveform capnography. Evaluate for chest rise and breath sounds.
10. Continuously monitor the airway with waveform capnography throughout transport. Reassess the position and function of the airway with every patient movement.

D. QuickTrach

1. Select appropriately sized device:

Pediatric patients < 2 years or < 10 kg: QuickTrach is contraindicated.
Needle cricothyrotomy may be performed if < 2 years or 10 kg.
Pediatric patients 10-35 kg (22-77 lb): QuickTrach Pediatric 2.0 mm
Adult patients >35 kg (77 lb): QuickTrach 4.0 mm

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2. Place the patient in a supine position with support under the shoulder and mild hyperextension of the head. If spinal precautions are indication, maintain the neck in a neutral position.
3. Locate the cricothyroid membrane.
4. Prepare the skin of the anterior neck with antiseptic solution.
5. Stabilize the cricoid cartilage with the nondominant hand.
6. Place the needle of the QuickTrach in the midline of the cricothyroid membrane and perforate the soft tissues at a 90-degree angle.
7. Aspirate for air return as the catheter is inserted. A saline flush with 2-3 ml of saline can be used in place of the included syringe.
8. Once air return is obtained, incline the QuickTrach device to a 45-degree angle and advance to the stopper.
9. Remove the stopper.
10. Hold the needle in place while advancing the catheter until the flange is against the skin. Remove the needle.
11. Secure with the padded strap.
12. Attach the connecting tube to the BVM to ventilate.
13. Confirm placement immediately with continuous waveform capnography. Evaluate for chest rise and breath sounds.
14. Continuously monitor the airway with waveform capnography throughout transport. Reassess the position and function of the airway with every patient movement.

E. Special Considerations

1. When this procedure is performed, a Quality Improvement report must be submitted to the EMS Agency within 24 hours. All cases where needle cricothyrotomy or QuickTrach are performed will undergo QA call review by the EMS Medical Director.
2. Risks of this procedure include damage to vessels and surrounding structures. Damage to the vocal cords may occur if the puncture is made too high. Perforation of the back wall of the trachea may occur if the puncture is too deep.
3. Needle cricothyrotomy is a temporizing measure to provide oxygenation until further advanced airway maneuvers can be performed in the hospital. Due to the small diameter of the catheter, it does not provide adequate ventilation and is meant to be used for a maximum of 30-45 minutes. Expect high end tidal CO₂ levels. Monitor for hyperinflation of the chest and if suspected, stop bagging to allow adequate time for exhalation.
4. A prolonged expiratory phase is necessary to allow adequate exhalation during ventilation through needle cricothyrotomy. A ratio of 1 second of inspiratory phase to 4 seconds of expiratory phase is recommended.